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Aragya Goyal

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Education

- **University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.)** **Pittsburgh, PA**
  B.S. - Computer Engineering (Autonomous Systems Focus); **GPA: 3.99/4.00** *August 2022 - April 2026*
- **Spring-Ford Area High School** **Royersford, PA**
  High School Degree (Summa Cum Laude); **GPA: 100.88/100.00** *August 2018 - June 2022*

Publications

- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, **A. Goyal,** A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, **"BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy,"** in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: (arxiv.org/abs/2505.07266)

Skills and Awards

- **Software**: Linux, ROS1, ROS2, Github, Python, C++, C, MATLAB, ARM Assembly, RISC-V Assembly, CANOpen, Google Colab
- **Electrical**: Altium, KiCAD, LTSpice, Soldering, Arduino, ESP32
- **Mechanical**: Solidworks, Onshape, Milling, Lathe, MIG Welding, Laser Cutting, General Shop Tools
- **Awards**: IEEE StuCon Micromouse Competition 3rd place, SSOE Design Expo 1st place Art of Making, FSAE Innovation Award, FIRST Chairman's Award, FIRST Excellence in Engineering, FIRST Industrial Design Award, VEX Judges Award, Eagle Scout

Professional Experience

- **Carnegie Mellon University Robotics Institute** **Pittsburgh, PA**
  Undergraduate Roboticist Researcher *April 2023 - Present*
  - **ZOË 2 Rover (Advisor: Dr. David Wettergreen)**:
    * Assisting in the development of software architecture and the robotic controller implementation for a novel 4-wheeled rover with a unique passive steering mechanism.
    * Enabling low-level software development for the rover using ROS2 and C++ to interface with motor controllers and encoders via CANOpen protocol.
    * Conducted motor and motor controller datasheet specifications validation via physical testing.
  - **Underwater Snake Robot (Advisor: Dr. Howie Choset)**: (youtu.be/tGJvrKFQcpM)
    * Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
    * Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
    * Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical/hardware maintenance.
  - **Apple's E-Waste Recycling Project (Advisor: Dr. Matthew Travers, Dr. Howie Choset)**:
    * Labeled datasets consisting of 4000+ images for YOLO Models to detect screws in iPhone images.
    * Integrated ROS1 and Python packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
    * Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.
- **ESTAT Actuation** **Pittsburgh, PA**
  Robotics Developer Co-op *May 2025 - November 2025*
  - Building and commissioning multiple robotic test stands expanding clutch testing capacity by 133%, covering mechanical assembly, electrical wiring, programming, and documentation for long-term use and potential sales to external customers.
  - Directing the assembly and delivery of 6 custom battery boxes for the Suzuki Moqba robot at the Japan Mobility Show. Overseeing quality control, leading debugging efforts to resolve heating issues, and ensuring a professional final build.
  - Designing and assembling customer facing demos for prototype versions of encoder and clutch integrated products.
  - Testing new generation rotary clutch prototypes to improve reliability and evaluate longevity. Creating documentation explaining testing procedures, and using these prototypes to inform design decisions for next generation production clutches.

Leadership & Activities

- **Robotics and Automation Society (University Rover Challenge)** **Pittsburgh, PA**
  Chief Engineer/Integration Lead *August 2022 - Present*
  - Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and bioscience teams.
  - Mentoring younger students in Onshape, KiCAD, and general manufacturing processes through hands-on building.
  - Securing and managing over $7,000 in funding and sponsorships to support team development and future growth.
  - Spearheaded the development of a prototype robotic hand utilizing pneumatics. (tinyurl.com/hydraarm)
- **University of Pittsburgh New Students Program** **Pittsburgh, PA**
  Panther Connect Leader *April 2024 - August 2024*
  - Selected as a campus ambassador to mentor incoming freshmen, providing guidance and support to facilitate a smooth transition to university life.
  - Organized and executed single-day and overnight orientation programs, ensuring engaging and well-organized experiences for new students.
- **Formula SAE - Panther Racing** **Pittsburgh, PA**
  R&D Engineer *August 2022 - March 2024*
  - Developed a 3D-printed e-brake bias system for a Formula SAE racecar using Solidworks, improving performance and usability; won the FSAE Innovation Award for implementation of the project. (tinyurl.com/ebrakeb)
  - Worked to design and code prototypes of slip-angle sensors which would allow for validation of slip angle using Raspberry Pi, mouse sensors, digital cameras, and IMUs.
  - Volunteered at the Pittsburgh Vintage Grand Prix, assisting with hill climb timing operations and public announcements.
- **FIRST Robotics** **Exton, PA**
  Team Captain/Design Lead *January 2020 - August 2022*
  - Led 40-student team to World Championships with highest win rate since 2005 and top 5% global ranking.
  - Designed award-winning competition robots and led R&D projects, including an experimental west-coast drive chassis, and a telescoping lift.
  - Placed 17th of 100 teams in weekend CADathon, designing a robot for weight manipulation and football launching.
- **Spring-Ford Technology Club (VEX Robotics)** **Royersford, PA**
  Vice President/Team Captain *August 2018 - June 2022*
  - Organized VEX robotics competition event for 60+ teams and received a scholarship for outstanding leadership within the club. (www.youtube.com/watch?v=iPC_ZToDl7I)
  - Designed, built, and coded 3 robots during my time, qualifying for the state level of competition each year.
  - Maintained an engineering notebook to track engineering decisions and progress through each season.
  - Mentored freshmen students on robot design and manufacturing through workshops and hands-on building.

Projects

- **Micromouse Robot**: Designed, assembled, and programmed a custom Micromouse robot through iterative PCB development, integrating motor control, IMU, IR sensors, and a display system with low-level embedded programming. Implemented PID control loops for navigation. Placed 3rd in competition at Stony Brook University IEEE conference. (tinyurl.com/goyalmm)
- **555 Timer Roulette Board**: Designed a PCB LED Roulette Wheel using 555 timer and CD4017 counter, featuring dynamic speed control using capacitive timing and a 3D-printed enclosure. (tinyurl.com/555roulette)
- **Bop It Project**: Developed a "Space Invaders Bop-It" game using a bare ATmega328P chip, integrating analog inputs, audio-visual feedback, and custom embedded logic for randomized commands and progressively challenging gameplay. (tinyurl.com/spacebopit)
- **VoltVitals Embedded Board**: Designed and prototyped an ESP32 based circuit with web interface to measure home outlet voltage and report over Wi-Fi, collaborating with FirstEnergy engineers to enable early outage detection. (tinyurl.com/voltvitals)
- **An Artistic take on Bipedal Robots**: Created a bipedal robot using Onshape and Arduino to carry a symbolic lantern given to all freshmen, reimagining a campus tradition through an artistic and technological lens. (tinyurl.com/gbipedal)
- **STM-32 Elevator Simulator**: Designed and implemented software architecture in ARM Assembly to operate a physical elevator simulator PCB. (tinyurl.com/stmelevator)
- **Custom Cane**: Designed and fabricated a walker-cane fusion using human-centered design methodologies to increase bathrooms accessibility for wheelchair users. Won first place at the Senior Design Expo. (tinyurl.com/CustomCane)